3790 - 7664 - 9428 (Andersbo Dyke)
- 7665 - 9429 (Gallsjön Dyke) Regards Sten-Åke

The Dala dolerites, central Sweden, and their palaeomagnetic signature

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New palaeomagnetic results are presented from nine dolerites from central Sweden. Data are also presented from four porphyry sites and one granite site. The data obtained are compared with previously published data from the area. Based on the directions of the characteristic magnetizations isolated in the various rock units a division into five groups, A—E, has been made. Group A consists of dolerites of late Sveconorwegian (c. 1050—850 Ma) age and group B comprises dolerites belonging to the Central Scandinavian Dolerite Group (c. 1250—1150 Ma). Groups C and D are less clearly defined. Group C, with a shallow negative northerly remanence inclination represents the Tuna dykes (c. 1370 Ma). The D direction, with a shallow positive northerly inclination, may represent an older generation of dolerites. The border between groups C and D is diffuse and may be the result of a prolonged mid-Proterozoic period with dolerite intrusions in Dalarna during which significant apparent polar wander occurred. Group E comprises dolerites with a steep positive inclination close to the direction of the Earth's present field. The porphyries and the granite studied carry group D and E remanence directions. This strengthens the evidence for a D-group generation of dolerites contemporary with the porphyries and the granite, c. 1650 Ma ago. □ *Precambrian palaeomagnetism, Fennoscandian APWP, dolerite, basalt, Dala sandstone, granite, Sveconorwegian, Jotnian, Subjotnian, Dalarna.*

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The Province of Dalarna, south central Sweden, is a region with a complex geology (Hjelmqvist 1966). Geological mapping, combined with geophysical measurements, carried out in its central part in connection with the Dala deep drilling project, has yielded new information of dolerites and other rocks in the area. In connection with a palaeomagnetic study of rocks related to the Siljan impact area (Elming & Bylund 1991) palaeomagnetic studies were made on a number of rock units outside the impact area. Bylund (1985) observed that some dolerites north of the Siljan region presented palaeomagnetic directions and palaeopoles that deviated from the established pattern and did not agree with published age data. This, and the presentation of the new apparent polar wander path (APWP) of Pesonen et al. (1989) which created a base for relative dating of the magnetization of Fennoscandian rock units, merited a new Palaeomagnetic study of the dolerites and other rocks in the province of Dalarna, particularly as the region appears as a key area for Fennoscandian palaeomagnetic studies. In this paper we discuss the palaeomagnetic results from dolerites in Dalarna and compare the data obtained with data from other rock units in the area.

General geology

The geology of Dalarna comprises rocks with ages that span from the early Proterozoic to the Silurian. The oldest units occur in the southern and eastern parts. They consist of acid metavolcanics, metasediments and intrusive granites of the early Proterozoic Svecofennian domain of the Fennoscandian Shield (Fig. 1). These rocks were reworked during the Svecofennian orogeny 1.93—1.78 Ga ago (Gaál & Gorbatschev 1987). This area is bordered in the west by the Transscandinavian Granite-Porphyry Belt (TGPB), stretching from southernmost Sweden north-northwestward until it is covered by the Caledonian nappes. The TGPB occupies large parts of central and western Dalarna and incorporates the Dala volcanics and the Dala granites.

The porphyries and granites of the TGPB in Dalarna comprise part of the Subjotnian complex. The Subjotnian supracrustals, volcanics (porphyries) and clastic sediments, were divided by Hjelmqvist (1966) into the Lower and Upper Dala Series, both intruded by the Dala granites. Porphyries in the eastern part give a Rb—Sr whole-rock age of 1634±38 Ma (Welin & Lundqvist 1970). Dala granite of the Järna-type has given Rb—Sr

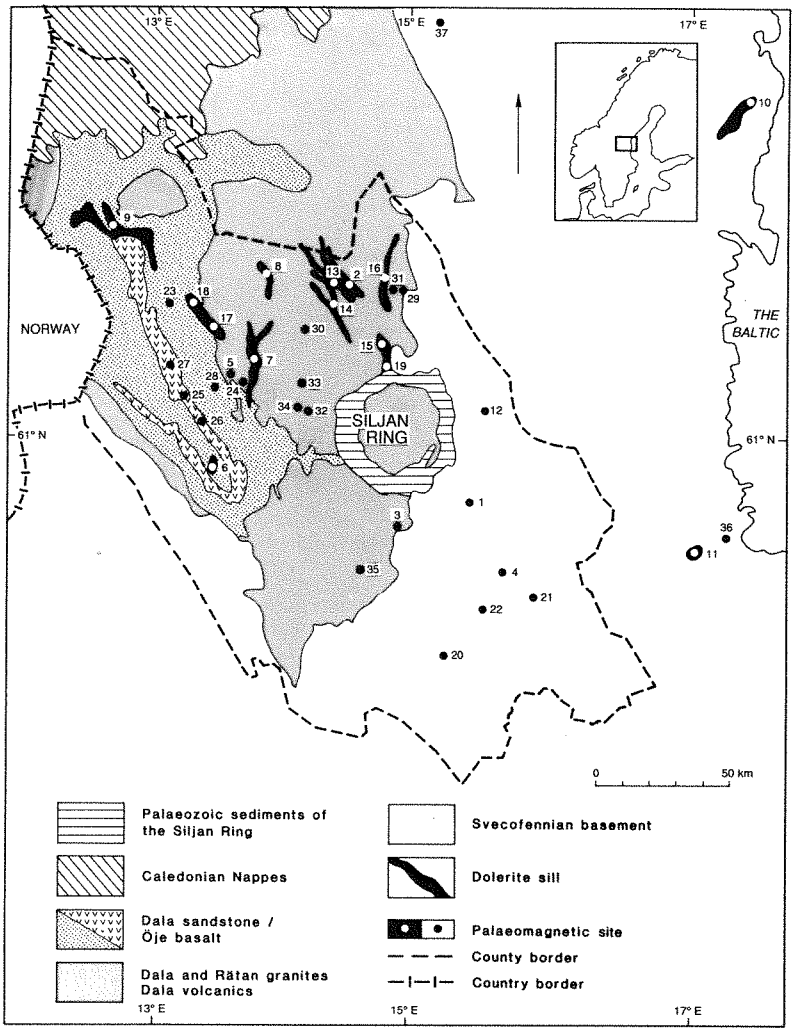


Fig. 1. Schematic geological map of Dalarna with palaeomagnetic sites.

whole-rock ages of 1675±12 and 1734±14 Ma (Åberg & Fredriksson 1984) and Rätan-type granite a Rb—Sr whole-rock age of 1649±25 Ma (Welin & Lundqvist 1977).

The Subjotnian complex is overlain by the Dala sandstone, a c. 800 m thick sandstone sequence intercalated with shales and conglomerates. The sandstone includes an effusive lava, the Öje basalt, that forms a c. 100 m thick conform layer within the sandstone. The age of the Dala sandstone and the Öje basalt is bracketed by the age of the underlying porphyries, c. 1630 Ma, and the age of intrusive dolerites, c. 1250 Ma. The Öje basalt yields K—Ar ages between 935 and 786 Ma (Priem et al. 1968). These ages are considered too young as both the basalt and sandstone are intruded by the 1250 Ma old dolerites. The Dala sandstone and the Öje basalt are included in the Jotnian complex. The c. 1250 Ma old dolerites of the Central Scandinavian Dolerite Group (CSDG) are sometimes considered a part of the Jotnian complex but are treated in this paper as a separate unit. This is substantiated by palaeomagnetic data.

The youngest rocks in Dalarna are the Ordovician and Silurian sediments, mostly limestones and sandstones, that are preserved in the Siljan Ring.

At least three generations of dolerites transect the bedrock of Dalarna. The youngest belong to the Blekinge-Dalarna Dykes (BDD), a set of dykes that are aligned parallel to the schistose border between the TGPB and the Southwest Scandinavian Gneiss Complex, known as the Protogine Zone. The BDD yield Rb—Sr and Sm—Nd intrusion ages ranging between 1050 and 850 Ma (Patchett 1978,

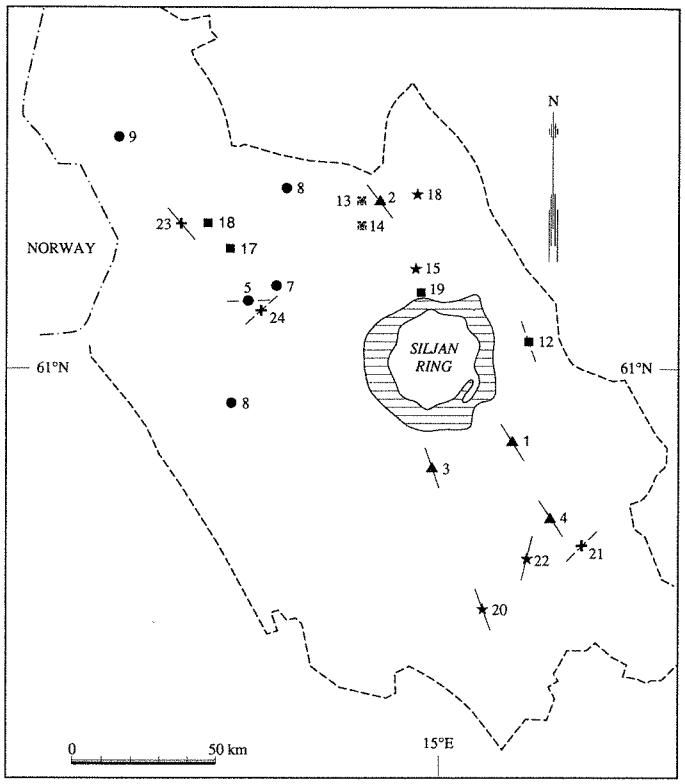


Fig. 2. Dolerite sites in Dalarna. Dykes are marked with lines indicating trend. Dashed line indicates that the trend is uncertain. The other sites are from sills with undefined orientation. The sites are marked according to group affiliation. Triangles denote Sveconorwegian dykes, group A, filled circles denote dolerites with CSDG magnetic orientations, stars denote group C dolerites, squares denote group D dolerites, combined squares and stars denote dolerites with both C and D components, crosses denote group E dolerites.

Johansson & Johansson 1990). They consist of subvertical, up to 100 m wide dykes trending north to nortwest, clearly observed on the aeromagnetic maps as discrete lineaments.

Next in age are the CSDG-dolerites. They occur in central Sweden and western Finland and were divided by Gorbatschev et al. (1979) into four complexes: the Satakunta complex in Finland, the Ulvö complex, the Jämtland complex, and the Dala complex in Sweden. They yield concordant K—Ar (Welin & Lundqvist 1975) whole-rock and mineral Rb—Sr ages (Patchett 1978) of 1250—1150 Ma. In Dalarna they occur mostly as flat-lying sheets.

In southern Dalarna a set of dolerites and quartz-porphyries (the Gustafs porphyries) that comprise the Tuna dykes (Hjelmqvist 1950) are found. Composite dykes also occur. These dykes have yielded a Rb—Sr whole-rock age of 1371 ± 50 Ma (Patchett 1978). They trend NNW to NNE and vary in width from a few cm to c. 15 m.

Åberg & Bollmark (1983) presented a Rb—Sr reference line of 1646 Ma for a clast of a composite dyke within the Siljan impact structure. The clast consists of an aplitic inner part and a doleritic outer part. The interpretation of this structure is that the aplite intruded the dolerite and that the age obtained is a minimum age for the dolerite. Patchett (1978) dated a dolerite in northern Dalarna (Bunkris, site 17 this paper) at 1516 Ma. He considered the isotopic system in the dyke as disturbed and had little confidence in the age obtained. But the palaeopole for the same dyke (Bylund 1985) indicated an age in agreement with the radiometric age. There are other dolerites in northeastern Dalarna that yield CSDG radiometric (Rb—Sr) ages but present deviating, probably older, palaeomagnetic directions (Bylund 1985). The conflicting data are enigmatic but the presence of a dolerite generation older than the Tuna dykes is possible.

The lack of contacts between the different units makes it difficult to evaluate the tectonics in the area. The Dala sandstones and the Subjotnian sediments in the sampled area are flat-lying with dips generally less than 15° and apparently the amount of tilt is minor. This includes the Öje basalt that is in concordance with the Dala sandstone. Fig. 2 shows dolerites represented by vertical to subvertical dykes in the southern half, while flat-lying sheets dominate in the northern half. The large dolerite sampled at site 7 (Älvdalsåsen) dips between 5 and 15° to the west (Hjelmqvist 1966), while the dolerite at site 16 (Älvho) dips 15—25° to

Table 1. Palaeomagnetic directions and palaeopoles for the Dalarna rock units. N = number of samples used for calculation of site means (*denotes calculation performed on site means). 95 = semi-angle of the 95% cone of confidence. k = the best estimate of Fisher's (1953) precision parameter. G = group affiliation. The ages given are Rb—Sr whole-rock and mineral ages after Patchett (1978). References: 1 = this paper, 2 = Multer (1971), 3 = Poorter (1976), 4 = Patchett & Bylund (1977), 5 = Piper & Smith (1980), 6 = Bylund (1985), 7 = Piper (1980).

Site, rock type	Slat. (°)	Slong. (°)	N	Decl. (°)	Incl. (°)	α95 (°)	k	G	Plat °N	Plong °E	Dp (°)	Dm (°)	Age Ma	Ref
1. Andersbo, D	60.76	15.41	7	143	42	9	39	A	1	229	6	9	—	1
2. Gällsjön, D	60.51	14.53	5	136	48	12	45	A	18	241	6	12	—	1
3. Ejen, D	60.67	14.79	8	128	24	9	45	A	6	248	5	9	—	6
4. Falun, D	60.59	15.58	10	132	46	6	75	A	−6	238	5	7	966 ± 20	4
5. Kvarnklitt, D	61.15	13.96	2	36	−48	—	—	B	5	163	—	—	—	1
6. Lybergsgnupen, ZDD, D	60.9	13.7	5	31	−42	10	62	B	1	166	8	12	—	2
7. Älvdalsåsen, D	61.28	13.79	11	50	−38	5	74	B	2	150	4	6	1231 ± 18	6
8. Mossi, D	61.55	13.85	5	45	−28	35	6	B	8	150	20	37	1217 ± 18	6
9. Jotnian dolerites, D	61.7	13.1	5*	37	−33	4	30	B	5	158	3	5	—	5
10. Gnarp, D	62.1	17.3	6	46	−52	8	78	B	12	159	7	11	—	3
11. Gävle, D	60.6	16.7	15	49	−24	4	10	B	8	149	2	4	—	3
12. Söndagsåsen, D	15.68	61.20	6	359	0	7	61	D	29	196	4	7	—	1
13a. Emåsjön, D	61.52	14.38	7	17	22	10	38	D	39	173	6	10	—	1
13b. Emåsjön, D	61.52	14.38	7	358	−34	7	104	C	10	196	4	8	—	1
14a. Harpsjön, D	61.48	14.37	6	32	15	3	643	D	31	157	1	3	—	1
14b. Harpsjön, D	61.48	14.37	6	356	−19	5	210	C	18	199	2	5	—	1
15. Emådalen, D	61.32	14.73	7	2	−3	16	16	C	27	193	8	16	1223 ± 23	6
16. Älvho, D	61.59	14.75	5	24	−16	12	41	C	19	190	6	12	1290 ± 63	6
17. Bunkris, D	61.43	13.48	8	357	2	4	245	D	27	197	2	4	1546 ± 84 1516 ± 63	6
18. Glysjön, D	61.51	13.32	6	11	17	8	68	D	37	180	4	8	—	6
19. Ämån, ZDA, D	61.2	14.8	5	28	4	7	138	D	27	163	4	7	—	2
20. Rämshyttan, D	60.25	15.25	6	10	−10	12	25	C	24	184	6	12	—	1
21. Fiskarbo, D	60.5	15.8	5	24	56	7	75	E	62	152	7	10	—	1
22. Tuna dykes, D	60.41	15.40	9	15	−18	10	30	C	21	180	5	10	1371 ± 50	6
23a. Nornäs, D	61.42	13.21	4	325	71	7	203	E	71	282	10	11	933 ± 13	6
23b. Nornäs, D	61.42	13.21	6	0	82	3	459	E	77	14	6	6	901 ± 39	5
24. Dysån, D	61.15	13.99	6	307	75	7	68	E	65	309	12	13	—	1
25. Gussjösätern, ZBB, B	61.0	13.4	6	8	8	5	195	D	33	184	3	5	—	2
26. Transtrand, ZBC, B	61.1	13.3	5	6	4	7	130	D	31	186	4	7	—	2
27. Jotnian lavas, B	61	13	8*	4	−2	11	28	D	28	189	5	11	—	5
28. Jotnian sediments, DS	61	13	6*	14	10	13	26	D	33	178	7	14	—	5
29. Noppikoski, P	61.50	14.84	10	12	17	19	6	D	33	178	7	14	—	1
30. Jöllen, ZSA, P	61.4	14.0	6	14	0	9	61	D	28	180	5	9	—	2
31. Älvbo, ZSD, P	61.50	14.75	5	20	−18	6	140	C	18	187	3	6	—	2
32. Gutsåvägen, P	61.10	14.16	2	40	52	—	—	E	52	133	—	—	—	1
33. Blyberget, P	61.16	14.16	5	343	65	6	106	E	73	236	8	10	—	1
34. Hållan, P	61.13	14.11	6	15	76	4	42	E	83	79	7	8	—	1
35. Dala Floda, G	60.53	14.75	9	3	54	7	42	E	64	189	7	10	—	1
36. Gävle granit, G	60.7	16.7	12	26	17	13	11	D	34	165	7	14	—	7
37. Turingen D	62.48	14.90	15	342	43	5	55	—	51	221	4	6	—	1

Table 2. Group mean palaeomagnetic directions and palaeopoles. Conventions as in Table 1.

Group	N	Decl. (°)	Inc. (°)	α95 (°)	k	Plat. °N	Plong. °E	Dp (°)	Dm (°)	Age of magnetization
A	4	134	40	10	46	−1	236	8	13	Late Sveconorwegian, 900—1000 Ma
B	7	42	−37	8	43	2	156	6	9	Early Sveconorwegian, 1200—1250 Ma
C	8	10	−15	9	32	21	184	5	9	Subjotnian-Jotnian, 1350—1650 Ma
D	13	12	13	8	21	35	180	4	8	Subjotnian-Jotnian, 1350—1650 Ma
E	8	4	69	10	25	82	178	14	17	

amples used for calcu-
% cone of confidence.
en are Rb—Sr whole-
, 3 = Poorter (1976),

)m °)	Age Ma	Ref
9	—	1
12	—	1
9	—	6
7	966 ± 20	4
—	—	1
12	—	2
6	1231 ± 18	6
37	1217 ± 18	6
5	—	5
11	—	3
4	—	3
7	—	1
10	—	1
8	—	1
3	—	1
5	—	1
6	1223 ± 23	6
2	1290 ± 63	6
4	1546 ± 84 1516 ± 63	6
8	—	6
7	—	2
2	—	1
0	—	1
0	1371 ± 50	6
1	933 ± 13	6
5	901 ± 39	5
3	—	1
5	—	2
	—	2
	—	5
	—	5
	—	1
	—	2
	—	2
	—	1
	—	1
	—	1
	—	1
	—	7
	—	1

1

, 900—1000 Ma
1, 1200—1250 Ma
350—1650 Ma
350—1650 Ma

the west. However, widely varying values for strike and dip have been reported and they suggest a lack of any preferred orientation (Lundqvist 1968). The dykes generally have the same trend across the province (Fig. 2). This fact and the shallow inclination of the sediments and the Öje basalt imply that no major tectonic movements have occurred since the end of the Svecofennian orogeny. However, minor local tilts and rotations cannot be excluded.

Palaeomagnetic measurements

Samples were collected either as drill-core or as blocks. In situ orientation was made with a magnetic compass or, when possible, with a suncompass.

NRM measurements of single rock samples were made on Digico and Molspin fluxgate magnetometers. In addition a number of specimens were measured on a SQUID-magnetometer at Stanford University, USA. Conventional Alternating Field (AF) and thermal demagnetization techniques were applied to isolate the different components of remanent magnetizations. After the thermal demagnetization steps the magnetic susceptibility was measured in order to check for alterations in the magnetic mineralogy. The remanence components were defined by a least square fit of vectors in two-dimensional plots (Torsvik 1986). Site mean directions were calculated according to the method of Fisher (1953). Results are listed in Tables 1 and 2. Figs. 3 to 5 present examples of remanence behaviour during demagnetization. Site mean directions are given in Fig. 6.

Dolerites

The AF-demagnetization of the dolerites showed a simple pattern with either a soft component removed a low peak fields (Fig. 3A) leaving a more stable component or a single component system (Fig. 3C).

The behaviour of the dolerites during thermal demagnetization varied from a single component system (Fig. 3B) with a narrow blocking temperature spectrum to a system where a soft low blocking temperature (Tb) component was removed early in the process leaving a hard characteristic component (Fig. 3D). Both types of behaviour could be found at the same site, and there was no correlation between age and pattern of behaviour. The microscopic study showed titanomagnetite and magnetite, partly martitized, as the main opaques. This magnetic assemblage is in agreement with the demagnetization tests.

Dala Series

Four sites of porphyries have been studied. One, site 29, is from the acid metavolcanics that form the base of the Subjotnian Dala volcanics while the other sites comprise volcanics of the upper part. A microscopic study of the volcanics showed magnetite and subordinate hematite as the main opaques. The coercivity (Hc) and blocking temperature spectrum of the rocks confirm the carriers of remanence as magnetite and hematite (Fig. 4). Specimens from site 29 (Fig. 4A—B) showed a gradual decrease during demagnetization indicating a wide range in Hc and Tb. The other sites (32, 33, 34) presented distinct hard Hc and high Tb components, probably carried by hematite and single domain magnetite (Fig. 4C—D).

One site (35) with granite of the Järna-type of the Dala Series presented demagnetization spectra comparable with those of the volcanics (Fig. 5).

Discussion

Five different groups of magnetizations (A—E) can be identified from the distribution of site mean directions and palaeopoles (Figs. 6, 7, 8). One group (A; Table 2) comprises data from dolerites of sites 1 to 4, the first two are from this study. The directions obtained are consistent with data from the BDD dolerites (Patchett & Bylund 1977, Bylund 1981, 1985) and it is clear that these sites represent late Sveconorwegian dykes belonging to the BDD-system.

Another distinct group (B; Table 2) is represented by sites 5 to 11. Site 5 is new and the other results are from previous studies (Mulder 1971, Poorter 1976, Piper & Smith 1980, Bylund 1985). These studies proved that the dykes belong to the Dala and Satakunta (site 11) complexes of the CSDG and, consequently, the dolerite represented by site 5 is suggested to belong to the same set of intrusions. Their age-range is c. 1250 to 1150 Ma.

Most of the dolerite sites 12 to 22 yielded NRM directions similar to those of the Subjotnian and Jotnian rocks presented below (Fig. 6) and their Virtual Geomagnetic Poles (VGP) are situated where the Subjotnian Loop overlaps itself (Fig. 8). It implies a magnetization older than that for the CSDG-dolerites, which contrasts to the Rb—Sr ages of 1215 and 1223 Ma obtained for two of the dolerites (sites 15 and 16; Patchett 1978). Another dyke, site 17, yielded a Rb—Sr age of 1516 Ma which is more in agreement with the palaeomagnetic data. But this dating was considered unreliable by Patchett (1978), who suggested that the dolerite

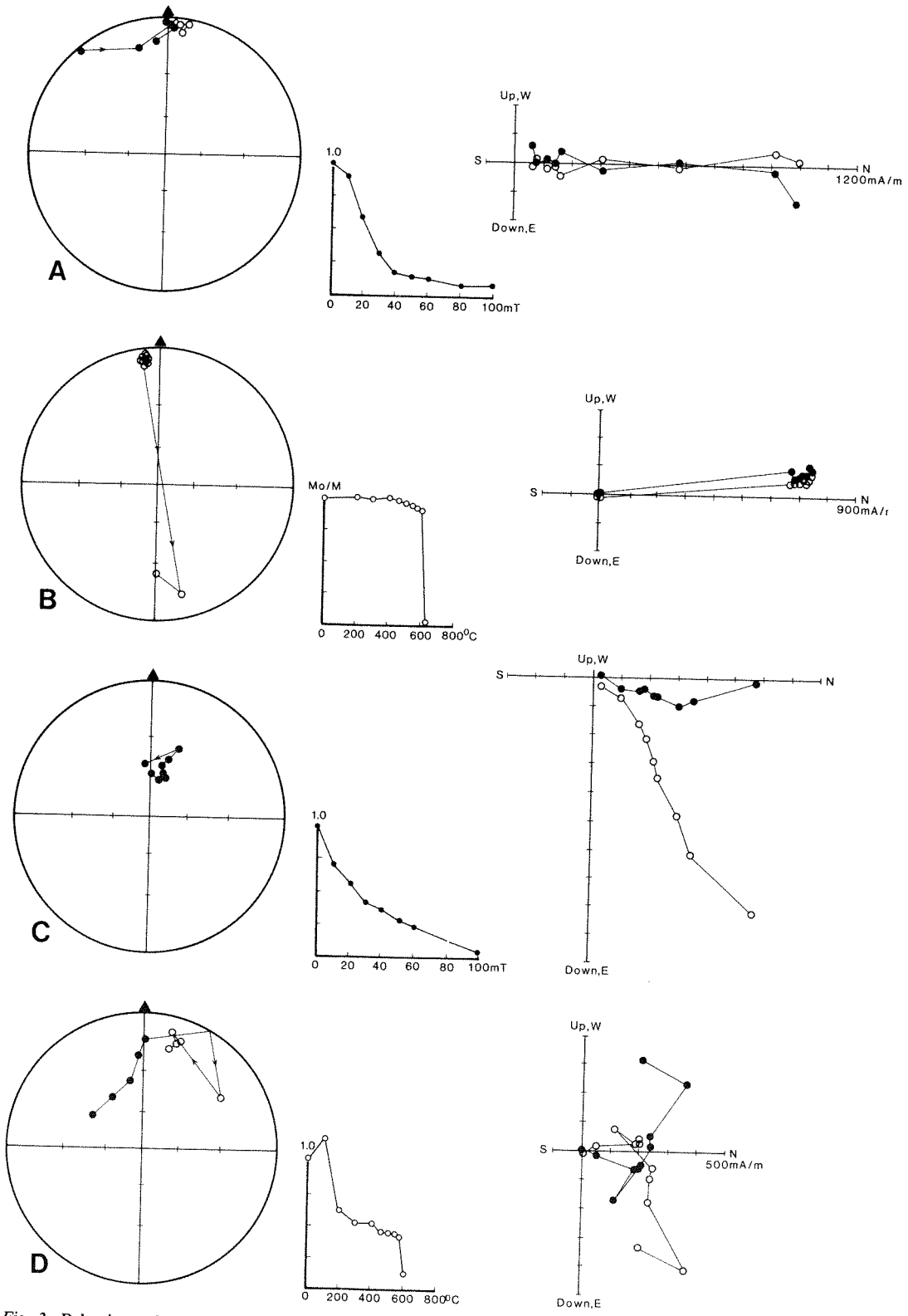
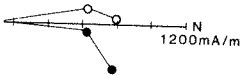


Fig. 3. Behaviour of remanent megnetization during progressive AF (A and C) and thermal (B and D) demagnetization. Dolerites. Left: stereoplot, closed (open) symbols denote positive (negative) inclination. Center: normalized curves for the magnetic intensity during demagnetization. Right: orthogonal vectordiagram, closed (open) symbols represent vector endpoints in the horizontal (vertical) plane.











and D) demagnetiza-
. Center: normalized
osed (open) symbols

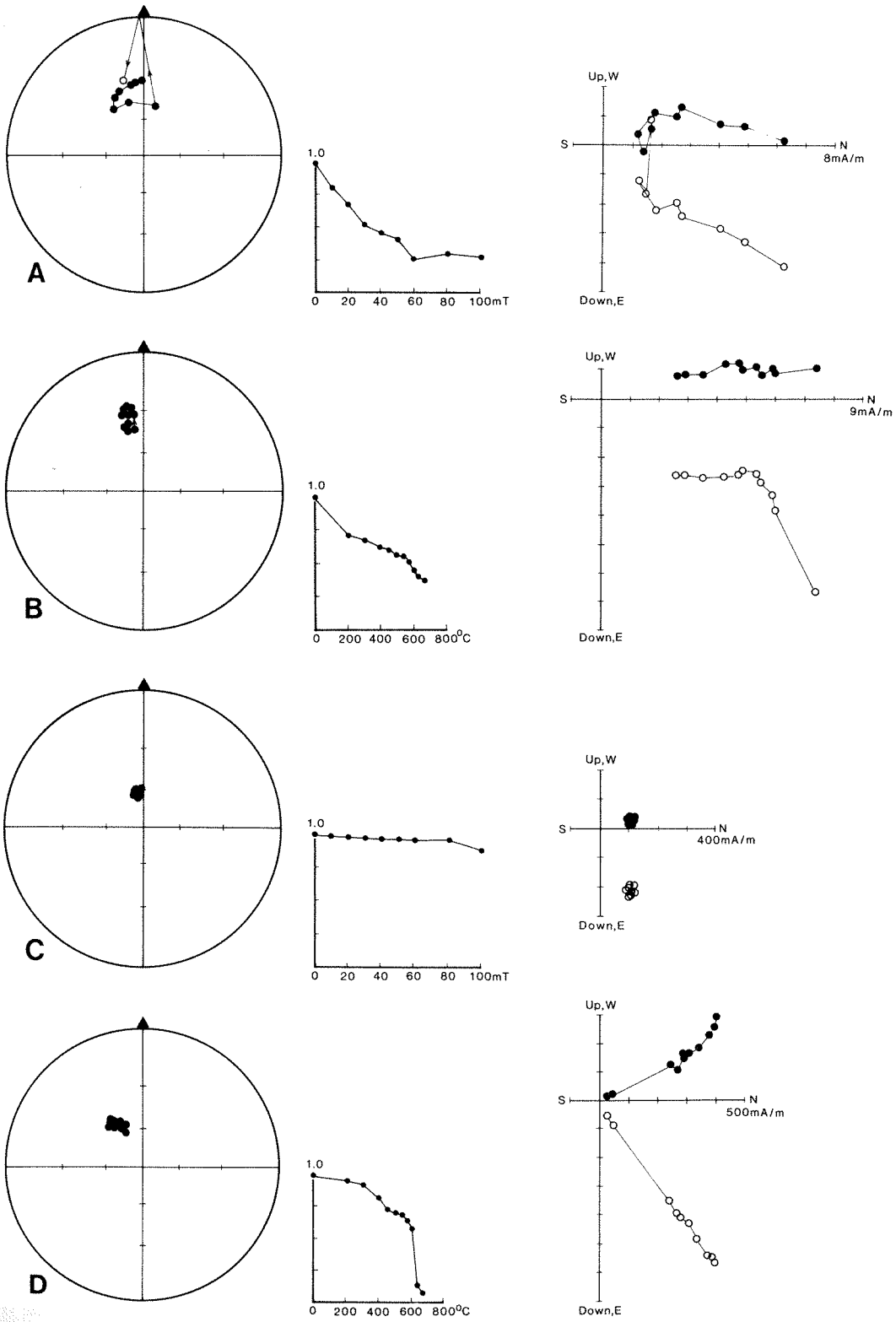


Fig. 4. Behaviour of the remanent magnetization during progressive AF demagnetization. Porphyries. Conventions as in Fig. 3.

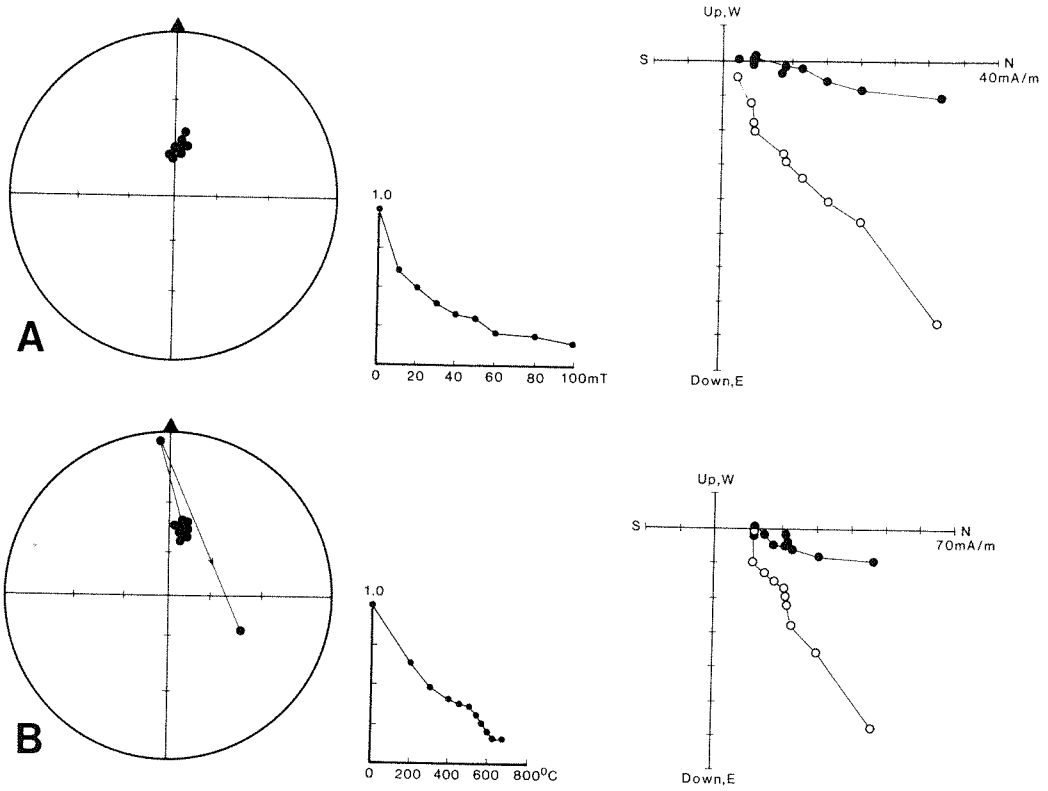


Fig. 5. Behaviour of the remanent magnetization during progressive AF and thermal demagnetization. Granite. Conventions as in Fig. 3.

was enriched in Sr which would present an age estimate that was too old.

Bylund (1985) suggested that the Siljan impact event at c. 360 Ma affected the NRM of the rocks in the areas surrounding the impact crater. However, Elming & Bylund (1991) show that the primary NRM in some of the rocks within the impact area survived the event. In this case the effects of the impact on the NRM of rocks situated as far away from the impact as the localities presented here should be insignificant.

The dolerites of sites 12 to 20 have been divided into two groups, C and D. This division is based on the differences in magnetic directions. Group C is represented by sites with northerly shallow negative inclination and group D by sites with northerly shallow positive inclination. The borderline between these two groups is however diffuse (e.g. sites 12, 15, 19; Table 1). Further, at sites 13 and 14 it is possible to separate the C and D components within the same specimens.

Sites 20 and 22 are from the southern part of Dalarna (Fig. 1 and 2) where the dolerite of the Tuna dykes is represented by site 22 and dated at 1371 ± 50 Ma (Rb—Sr; Patchett 1978). Apparently these dykes belong to the same C-generation. In the northern part of the province sites 15 and 16 have distinct group C directions while sites 13 and 14 carry both a C and D component.

Group D is represented by sites 12, 17—19 (Table 2). Further, sites 13 and 14 have D-components isolated by both AF and thermal demagnetization in Tb and Hc ranges lower than the C component. However, the occurrence of secondary fine-grained magnetite in this dolerite implies that the C component is a CRM and younger than the D component. The age of the dolerite generation carrying the D magnetization is enigmatic. The ages of 1646 and 1516 Ma, presented by Åberg & Bollmark (1983) and Patchett (1978), are too vaguely defined to design an age to the D-group dolerites. Another possibility is that the C and D groups represent a prolonged dolerite intrusion event during which significant polar wander occurred. The existence of an older dolerite generation in the vicinity of Dalarna is given by data from site 37 (Fig.

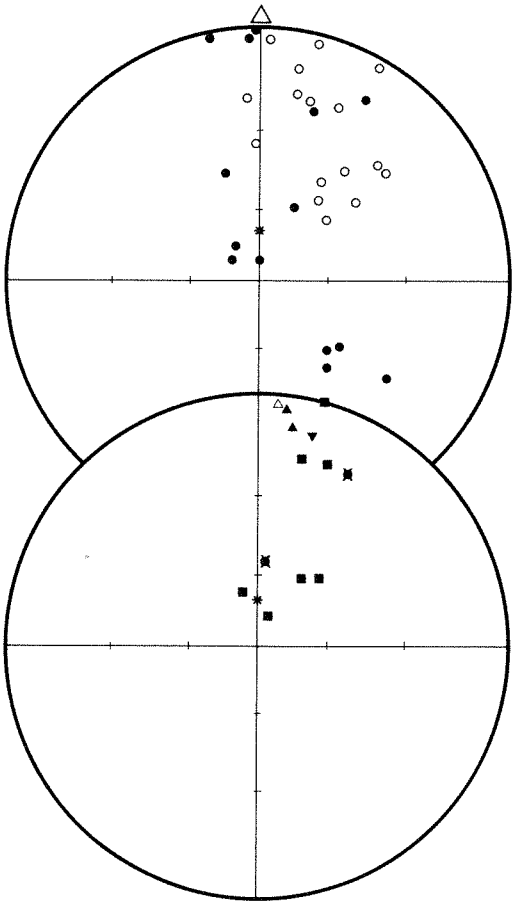


Fig. 6. Site mean directions of characteristic magnetizations. Data from Table 1. Star denotes the Earth's present field. Stereoplot, closed (open) symbols denote positive (negative) inclination. Top, dolerites (circles); bottom, basalt (triangle), Dala sandstone (inverted triangle), porphyry (square), granite (crossed circle).

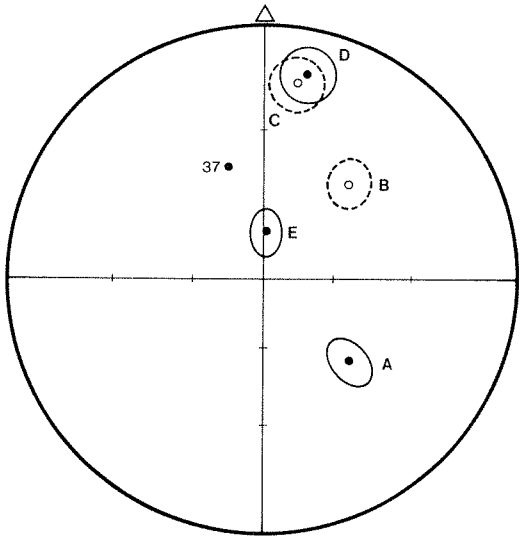


Fig. 7. Group mean directions with 95% confidence limit circles. Conventions as in Fig. 6. Note that the open symbols plot on the upper hemisphere, the closed on the lower hemisphere.

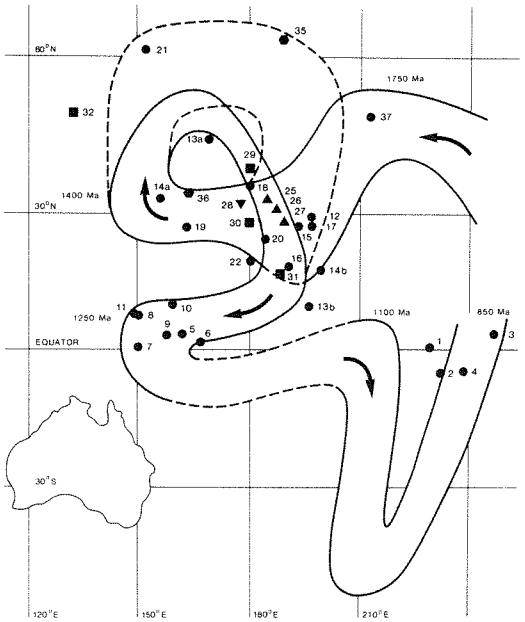


Fig. 8. The APWP after Pesonen et al. (1989) with the Dalarna palaeopoles from Table 1. The dashed hump implies a possible extension of the Subjotnian Loop.

1; Table 1) situated north of the province (Bylund, prel. data). The corresponding VGP corresponds to an age of c. 1750 Ma on the Fennoscandian APWP. However, the remanence directions of site 37 do not fit any of the groups presented.

Group E (Table 2) consists of sites 21, 23, and 24 which yield steep positive inclinations. Site 23 is dated at 901—933 Ma (Patchett 1978; Rb—Sr) and, accordingly, it belongs to the Sveconorwegian BDD dykes. The data in 23a are from Bylund (1985) and 23b from Piper & Smith (1980) and the rocks studied were sampled at approximately the same site. Their magnetic directions may be compared with directions found in the porphyries at sites 33 and 34, presented below.

The results from the Subjotnian and Jotnian rocks present a rather obscure picture with a majority of shallow northerly directions with both negative (C) and positive (D) inclinations and the rest with a steep positive (E) inclination (Table 2). Those with shallow inclination give palaeopoles that fall in the overlapping area of the Subjotnian Loop of the Fennoscandian APWP (Fig. 8) while

the others fall outside the APWP. The data are from granite, porphyry, basalt, and Dala sandstone. The ages range from c. 1650 Ma for the Subjotnian granites and porphyries while the Jotnian sandstones and basalts have ages bracketed to between 1630 and 1250 Ma. The Subjotnian Loop (Fig. 8) starts with a number of palaeopoles from dolerites and Rapakivi granites in Finland (c. 1600 to 1550 Ma, Pesonen et al. 1989). The Loop itself is based on data from a set of Rapakivi granites with associated anorthosites and gabbros from central Sweden including the Gävle granite, site 36 (Table 1). The ages reported for these rock units range from c. 1550 Ma (Rb—Sr) for the Nordingrå granite and anorthosite-gabbro to c. 1290 Ma (Rb—Sr) for the Ragunda massifs (for a discussion and references on the ages and palaeopoles of the rocks that constitute the Subjotnian Loop see Bylund 1985).

The Dala porphyries and granite represented by sites 29—31 and 36 have palaeopoles in the overlapped, early part of the APWP, in accordance with their age. Of the porphyries and granite sites with steep positive inclination, sites 32 and 35 yield palaeopoles in the vicinity of the established APWP (Fig. 8) while sites 33 and 34 yield poles at high latitudes well outside the APWP (not shown in Fig. 8). Bylund (1985) discussed a similar case from the E—W trending Hällefors and Breven dykes and the related Eskilstuna dolerites. A hump was suggested on the Subjotnian Loop to incorporate poles with directions like the poles found at sites 32 and 35 of this study. A similar hump is sketched in Fig. 8. An alternative is that it was not possible with the available techniques to resolve the different components of a multicomponent magnetic system in the rocks studied. Sites 33 and 34 have directions close to the Earth's present field (Fig. 6) and a viscous origin of that magnetization cannot be excluded. However, the porphyry of site 33 presents a magnetization with high Tc and high Tb (Fig. 4C—D) and it is difficult to find a recent event that could have imprinted such a stable NRM. There is always the possibility that Fennoscandia was at a similar latitude as at present sometime in the middle Proterozoic, but there is no evidence in the available palaeomagnetic record for this.

The Jotnian Dala sandstone and Öje basalt (sites 25—28) yield remanence directions similar to those found in the Subjotnian porphyries and granite with VGPs in the overlapping part of the Subjotnian Loop (Fig. 8; Table 1).

Nyström (1982, 1983) studied rocks in an area north and west of the Siljan impact structure and showed that the Subjotnian and Jotnian units have undergone metamorphism from prehnite-pumpellyite to pumpellyite-actinolite facies. This is a low temperature process, up to a few hundred degrees C, but if it has a long duration it may affect the NRM of the rocks. Nyström further showed that only the Subjotnian and Jotnian rocks and not the CSDG-dolerites were affected by this metamorphism. It may also have affected the pre-CSDG-dolerites present in the area and explain the secondary CRM component found at sites 13 and 14 and the similarities in magnetic directions between some of the dolerites, e.g. sites 12, 15—19 and the shallow Subjotnian and Jotnian directions observed (Table 1; Fig. 6 and 7).

The rocks presented here have normal magnetic polarity according to the definition given by Pesonen & Neuvonen (1981) with the exception of the four Sveconorwegian dolerites which have reversed polarity. However, Elming (1985 and in prep.) reports Subjotnian reversed polarities from basic rocks in the northern part of Sweden, which means that the Subjotnian was a period with mixed polarities. It also indicates that the scatter of the NRMs observed in the group C and D rocks are due to partial or complete remagnetization due to metamorphism during a time interval dominated by positive polarity.

Conlusion

The results from this study of 14 sites, comprising dolerites, porphyries, and a granite, combined with previously obtained data, further prove the existence of Sveconorwegian and CSDG-dolerites in the province of Dalarna. They also prove that the Tuna dykes may be spread over a larger area than hitherto expected. The existence of a dolerite generation older than the Tuna dykes is implied. The determination of the ages of the NRM of the Subjotnian and Jotnian rock sequences is obscured by the fact that the majority of the palaeopoles fall in the area where the Subjotnian Loop of the Fennoscandian APWP overlaps itself. An interpretation with the Subjotnian palaeopoles in the overlapped part and the Jotnian in the overlapping part is assumed. However, a partial or complete remagnetization of the group C and D rocks during Jotnian time cannot be excluded. The significance of the NRM found in the group E rocks with steep positive inclination is still unclear.

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